How Empathy Appeals vs. Economic Incentives Shift Negotiation and Pricing Strategies in LLMs

Experimental Approach and Human Data Collection Plan

This document lays out, step by step, how the project will be run: what is being tested, what humans will actually be asked to do, what data is collected from them, how many participants are needed, and how the results will be analyzed.

1 Research Design

1.1 Research Question

Does the type of persuasive appeal a human uses — **empathy-based** or **economic/instrumental** — change how an LLM negotiates and prices a deal?

1.2 Design Type

This is a **between-subjects experiment**: each participant is randomly placed into only *one* condition, so we compare different groups of people rather than asking the same person to try both styles.

1.3 Conditions

- **Condition A — Empathy Appeal:** the participant is told to negotiate using personal circumstances, goodwill, and relationship-building language (e.g., explaining why the discount matters to them).
- **Condition B — Economic/Instrumental Appeal:** the participant is told to negotiate using rational, self-interest-based arguments (e.g., market price comparisons, bulk/loyalty discounts, or the option to buy elsewhere).
- **Condition C — Neutral/Control (optional):** the participant negotiates with no specific style instruction, giving a baseline to compare both styles against.

1.4 What We Measure (Dependent Variables)

- **Final negotiated price** — the main outcome of interest.
- **Concession rate** — how much the LLM reduced its price relative to where it started.
- **Number and size of concessions**, and **how many turns** it took to reach a deal.
- **Whether a deal was reached at all.**
- The LLM's **tone and behavioral pattern** (e.g., firm, accommodating, reciprocating).

1.5 Participant-Level Covariates

Since people differ naturally, we also record a few background traits to make sure any differences we see are really caused by the appeal type, and not just by who happened to land in which group: **trait empathy**, **trust in AI**, **prior negotiation experience**, and **prior familiarity with LLMs**.

1.6 Keeping the LLM Constant

For the comparison to be fair, everything about the LLM must stay exactly the same across every single participant and condition — the same model, the same seller persona and system prompt, the same starting price, the same generation settings, and the same turn limit. The only thing allowed to differ between participants is the assigned appeal style. We also give the LLM a **hidden reservation price** — the lowest price it is allowed to accept, invisible to the participant — so the negotiation has real stakes instead of the LLM simply giving the item away.

1.7 Hypotheses

- **H1:** Empathy appeals will lead to a *lower final price* and *larger concessions* than economic appeals.
- **H2:** The two appeal styles will differ not just in the final price, but in *when* concessions happen during the conversation (e.g., empathy appeals producing earlier, gradual softening).
- **H3:** The effect of appeal type on final price will still hold even after accounting for each participant's empathy, trust, and experience levels.

2 Materials and Participant Flow

2.1 What Needs to Be Built First

A single fixed negotiation scenario (e.g., "negotiate the price of a used laptop listed at $500"); a fixed LLM system prompt containing the seller persona and hidden reservation price; two (or three) short instruction scripts, one per condition; a chat interface that automatically saves every message as it happens; a short pre-survey and post-survey; and a brief plain-language note telling participants what the study involves.

2.2 What Each Participant Actually Does

1. Reads a short note about the study and agrees to take part.
2. Fills a quick pre-survey: basic demographics, a short empathy scale, and a short AI-trust scale (∼2 minutes).
3. Is randomly and invisibly placed into a condition by the system.
4. Reads their condition's instructions on how to frame their negotiation.
5. Chats live with the LLM to try to get the best price, up to about 10 back-and-forth turns (∼5–10 minutes).
6. Fills a short post-survey about fairness, satisfaction, and what kind of argument they felt they used (∼3 minutes).

Total time per participant: roughly 10–15 minutes.

3 Human Data Requirements

This is exactly what we need to collect from every participant, split into four stages:

- **Before the task:** age, gender, education, prior negotiation experience, prior LLM usage, trait-empathy score, AI-trust score, and their (hidden) condition assignment.
- **During the task (collected automatically):** the complete, timestamped chat transcript; the

price offered at every turn; the final agreed price; how many turns it took; and whether a deal was reached at all.

- **After the task:** a manipulation check asking what kind of argument they felt they used; how fair they found the outcome; how satisfied they were with the LLM; and how effective they think their own strategy was (all on simple 1–7 rating scales).

Field	Example
Participant ID / Condition	P0042 / Empathy
Chat transcript	[buyer msg 1, LLM msg 1, buyer msg 2, . . .]
Turns / Final price / Deal reached	6 turns / $420 / Yes
First concession / Concession rate	$20 / 16%
Pre-survey scores	Trust = 4.2, Trait empathy = 5.1
Post-survey scores	Fairness = 3, Satisfaction = 4

Table 1: What one row of the final dataset looks like (one row per participant)

4 Sample Size

The target is **100–200 participants** in total. With two conditions, that works out to roughly 50–100 people per group; with three conditions, roughly 35–70 per group. As a rule of thumb, detecting a moderate difference between groups reliably (what statisticians call a "medium effect size," $d \approx 0.5$) needs about 64 people per group at conventional confidence levels — so this range gives enough participants with some buffer (10–15%) left over for anyone who drops out, fails an attention check, or doesn't finish the chat.

5 Analysis Plan

5.1 Comparing the Groups

The main comparison is the final negotiated price between conditions, using a t-test (two conditions) or ANOVA (three conditions), reporting group averages, confidence intervals, and effect size. If there are three conditions, follow-up pairwise tests (Tukey's HSD) show exactly which pair(s) of conditions differ.

5.2 Looking Deeper Than Just the Final Price

We also compare the *concession rate* (how much of the starting price was given up):

$$\text{Concession Rate} = \frac{\text{Starting Price} - \text{Final Price}}{\text{Starting Price}}$$

along with how many turns each negotiation took and whether a deal happened at all. A regression model also checks whether the appeal-type effect survives after accounting for each participant's empathy, trust, and experience — this tells us whether the effect is really about the appeal style, or just about who happened to be more persuasive in general.

5.3 Reading the Conversation Itself

Beyond the final numbers, we look at the shape of the negotiation: did the LLM concede early and gradually, or hold firm and only concede late? We also read through transcripts to tag recognizable behaviors — acknowledging the buyer, making a counteroffer, holding firm, or reciprocating a concession — so we can describe *how*, not just *how much*, the LLM's behavior changed.

5.4 Making Sure the Experiment Actually Worked (Validity Checks)

These checks exist to catch a specific risk: what if a participant assigned to the "empathy" condition doesn't actually write empathetic messages, or the LLM doesn't behave consistently? If that happens, any pattern we find wouldn't really be about empathy vs. economics at all — it would just be noise. Two checks guard against this:

- **Did participants follow their assigned style?** We ask them directly in the post-survey, but self-reports can be unreliable, so we also read the actual chat messages ourselves (or with a friend helping) and independently judge whether the language used matches the assigned condition, without looking at which condition they were assigned to first (this avoids bias). If most of a participant's messages don't match their assigned style, that session is flagged and can be excluded or analyzed separately.
- **Did the LLM behave consistently?** Before collecting data from all 100–200 participants, we run a small trial of about 15–20 test conversations across both conditions and check three things: the LLM never agrees to a price below its hidden minimum, it actually negotiates back and forth instead of instantly accepting or refusing, and it stays in character as the same seller every time. If it fails any of these, we fix the prompt and re-test before running the real study — this avoids discovering a broken setup only after collecting all the real data.

6 Summary Pipeline

Design the experiment → Build materials → Small pilot test → Recruit 100–200 participants → Consent + Pre-survey → Random condition assignment → Human–LLM negotiation (auto-logged) → Post-survey → Check transcripts match conditions → Final dataset (100–200 rows) → Statistical & qualitative analysis